

EXPERIMENT 7

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EXPERIMENT: Advanced SQL Queries

AIM: To study and analyze functional dependencies and normalization of relational databases by identifying candidate keys, prime and non-prime attributes, determining the highest normal form, and decomposing relations to reduce redundancy and improve database design.

Question 1: 1. R(ABCDE)

FDs

- $CE \rightarrow D$
- $D \rightarrow B$
- $C \rightarrow A$

Answer: Candidate Key-

Attributes C and E never appear on the RHS, so they must be in every key.

Calculate:

$(CE)^+$

$= C, E$

$C \rightarrow A \rightarrow A$

$CE \rightarrow D \rightarrow D$

$D \rightarrow B \rightarrow B$

Therefore:

$(CE)^+ = \{A, B, C, D, E\}$

So:

Candidate Key = CE

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Prime Attributes

C, E

Non-Prime Attributes

A, B, D

Normal Form

- $CE \rightarrow D$: full dependency on the complete key.
- $D \rightarrow B$: D is not a superkey, and B is non-prime.

So 3NF is violated.

There is no partial dependency of a non-prime attribute on C or E alone.

Hence, Highest Normal Form = 2NF

Candidate Key = {CE}

Question 2: R(ABCDEF)

FDs

- $AB \rightarrow C$
- $DC \rightarrow AE$
- $E \rightarrow F$

Answer: Candidate Keys

$(ABD)^+$

=A,B,D

$AB \rightarrow C$

$DC \rightarrow A, E$

$E \rightarrow F$

Therefore:

$(ABD)^+ = \{A, B, C, D, E, F\}$

So ABD is a candidate key.

$(BCD)^+$

=B,C,D

$DC \rightarrow A, E$

$E \rightarrow F$

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Therefore:

$(BCD)^+ = \{A, B, C, D, E, F\}$

So BCD is also a candidate key.

Candidate Keys:

ABD and BCD

Prime Attributes: A, B, C, D

Non-Prime Attributes: E, F

Normal Form

Consider:

$DC \rightarrow E$

Since DC is a proper subset of candidate key BCD, and E is non-prime:

$DC \rightarrow E$ is a partial dependency.

Therefore, the relation violates 2NF.

Hence, Highest Normal Form = 1NF

Candidate Keys = $\{ABD\}$, $\{BCD\}$

Question 3: R(ABCDEFGHI)

FDs

- $AB \rightarrow C$
- $BD \rightarrow EF$
- $AD \rightarrow GH$
- $A \rightarrow I$

Answer: Attributes A, B, D do not appear on the RHS of any FD, so they must be included in the key.

Calculate:

$(ABD)^+$

= A, B, D

$AB \rightarrow C$

$BD \rightarrow E, F$

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$AD \rightarrow G, H$

$A \rightarrow I$

Therefore:

$(ABD)^+ = \{A, B, C, D, E, F, G, H, I\}$

Thus:

Candidate Key = ABD

Prime Attributes: A, B, D

Non-Prime Attributes: C, E, F, G, H, I

Normal Form

There are several partial dependencies:

- $AB \rightarrow C$
- $BD \rightarrow EF$
- $AD \rightarrow GH$
- $A \rightarrow I$

All of these have determinants that are proper subsets of candidate key ABD, and their RHS attributes are non-prime.

Therefore, 2NF is violated.

Hence, Highest Normal Form = 1NF

Candidate Key = {ABD}

Question 4: R(ABCDE)

FDs

- $AB \rightarrow CD$
- $D \rightarrow A$
- $BC \rightarrow DE$

Answer: $(AB)^+$

$AB \rightarrow C, D$

Now we have B, C:

$BC \rightarrow D, E$

Therefore:

$(AB)^+ = \{A, B, C, D, E\}$

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So AB is a candidate key.

$(BC)^+$

$BC \rightarrow D, E$

$D \rightarrow A$

Therefore:

$(BC)^+ = \{A, B, C, D, E\}$

So BC is a candidate key.

$(BD)^+$

$D \rightarrow A$

Now A, B are available:

$AB \rightarrow C, D$

Then $BC \rightarrow E$

Therefore:

$(BD)^+ = \{A, B, C, D, E\}$

So BD is also a candidate key.

Candidate Keys: AB, BC, BD

Prime Attributes: A, B, C, D

Non-Prime Attribute: E

Normal Form

Check:

$D \rightarrow A$

D is not a superkey, because D alone cannot determine all attributes.

However, A is a prime attribute.

Therefore, this FD satisfies 3NF but violates BCNF.

The other dependencies do not violate 3NF.

Hence, Highest Normal Form = 3NF

Candidate Keys = $\{AB\}$, $\{BC\}$, $\{BD\}$

Question 5: R(ABCDE)

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FDs

- $BC \rightarrow ADE$
- $D \rightarrow B$

Answer: $(BC)^+$

$BC \rightarrow A, D, E$

Therefore we get:

A, B, C, D, E

So:

BC is a candidate key.

$(CD)^+$

$D \rightarrow B$

Now B,C are available:

$BC \rightarrow A, D, E$

Therefore:

$(CD)^+ = \{A, B, C, D, E\}$

So CD is also a candidate key.

Prime Attributes: B, C, D

Non-Prime Attributes: A, E

Normal Form

Consider:

$D \rightarrow B$

D is not a superkey.

However, B is a prime attribute because B occurs in candidate key BC.

Therefore, the FD satisfies 3NF.

But it violates BCNF because D is not a superkey.

Hence, Highest Normal Form = 3NF

Candidate Keys = $\{BC\}, \{CD\}$

Question 6: R(ABC)

FDs

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- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow A$

Answer: A+

$A \rightarrow B$

$B \rightarrow C$

Therefore:

$A^+ = \{A, B, C\}$

So A is a candidate key.

B+

$B \rightarrow C$

$C \rightarrow A$

Therefore:

$B^+ = \{A, B, C\}$

So B is a candidate key.

C+

$C \rightarrow A$

$A \rightarrow B$

Therefore:

$C^+ = \{A, B, C\}$

So C is a candidate key.

Candidate Keys: A, B, C

Prime Attributes: A, B, C

Normal Form

For every FD:

- $A \rightarrow B$: A is a candidate key.
- $B \rightarrow C$: B is a candidate key.
- $C \rightarrow A$: C is a candidate key.

Therefore, every determinant is a superkey.

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Hence, the relation satisfies BCNF.

Hence, Highest Normal Form = BCNF

Candidate Keys = {A}, {B}, {C}

Question 7: Consider a relation P having fields F1, F2, F3, F4, F5 with the functional dependencies:

- $F1 \rightarrow F3$
- $F2 \rightarrow F4$
- $(F1, F2) \rightarrow F5$

Answer: Step 1: Find the Candidate Key

Consider {F1, F2}.

Calculate its closure:

$(F1, F2)^+$

$= \{F1, F2\}$

Using $F1 \rightarrow F3$:

$\rightarrow F3$

Using $F2 \rightarrow F4$:

$\rightarrow F4$

Using $F1F2 \rightarrow F5$:

$\rightarrow F5$

Therefore:

$(F1, F2)^+ = \{F1, F2, F3, F4, F5\}$

Hence, the candidate key is:

Candidate Key = {F1, F2}

Step 2: Prime and Non-Prime Attributes

Prime attributes: F1, F2

Non-prime attributes: F3, F4, F5

Attribute	Type
F1	Prime
F2	Prime
F3	Non-Prime

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F4	Non-Prime
F5	Non-Prime

Step 3: Determine the Normal Form

The relation is assumed to satisfy 1NF because its attributes contain atomic values.

Now check for 2NF.

Candidate key = F1F2

But:

$F1 \rightarrow F3$

Here, F1 is only a part of the candidate key F1F2, and F3 is non-prime.

So this is a partial dependency.

Similarly:

$F2 \rightarrow F4$

F2 is also only a part of candidate key F1F2, and F4 is non-prime.

Therefore, the relation violates 2NF.

Highest Normal Form = 1NF

Final Answer-

- Candidate Key: {F1, F2}
- Prime Attributes: F1, F2
- Non-Prime Attributes: F3, F4, F5
- Highest Normal Form: 1NF
- Reason: $F1 \rightarrow F3$ and $F2 \rightarrow F4$ are partial dependencies of non-prime attributes on parts of the composite candidate key.

Decomposition to remove redundancy

If decomposition is required:

1. P1(F1, F3) — $F1 \rightarrow F3$
2. P2(F2, F4) — $F2 \rightarrow F4$
3. P3(F1, F2, F5) — $F1F2 \rightarrow F5$

So, 3 tables are formed after decomposition.

Question 8: Consider the relation:

R(A, B, C, D, E, F, G, H, I)

Functional dependencies:

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- $AB \rightarrow C$
- $AD \rightarrow GH$
- $BD \rightarrow EF$
- $A \rightarrow I$
- $H \rightarrow J$

Answer: $AB \rightarrow C$

$AD \rightarrow GH$

$BD \rightarrow EF$

$A \rightarrow I$

Notice that A, B, and D do not appear on the RHS of any applicable FD.
Therefore, they must be included in the candidate key.

$(ABD)^+$

Initially:

$\{A, B, D\}$

Using $AB \rightarrow C$:

$\rightarrow C$

Using $AD \rightarrow GH$:

$\rightarrow G, H$

Using $BD \rightarrow EF$:

$\rightarrow E, F$

Using $A \rightarrow I$:

$\rightarrow I$

Therefore:

$(ABD)^+ = \{A, B, C, D, E, F, G, H, I\}$

Since the closure contains all attributes:

Candidate Key = $\{A, B, D\}$

Step 2: Prime and Non-Prime Attributes

Prime Attributes: A, B, D

Non-Prime Attributes: C, E, F, G, H, I

Attribute	Type
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A	Prime
B	Prime
C	Non-Prime
D	Prime
E	Non-Prime
F	Non-Prime
G	Non-Prime
H	Non-Prime
I	Non-Prime

Step 3: Determine the Highest Normal Form

Candidate key:

ABD

Now check the dependencies.

1. $AB \rightarrow C$

AB is only a part of the candidate key ABD.

And C is a non-prime attribute.

Therefore:

$AB \rightarrow C$ is a partial dependency.

So, 2NF is violated.

2. $AD \rightarrow GH$

AD is also a proper subset of ABD.

G and H are non-prime.

Therefore, this is also a partial dependency.

3. $BD \rightarrow EF$

BD is a proper subset of ABD.

E and F are non-prime.

Again, this is a partial dependency.

4. $A \rightarrow I$

A is a proper subset of candidate key ABD, and I is non-prime.

Therefore, this is also a partial dependency.

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Hence:

Highest Normal Form = 1NF

The relation is not in 2NF because several non-prime attributes depend on only part of the composite candidate key.

Step 4: Decomposition

We separate the functional dependencies into individual relations.

Table 1

R1(A, B, C)

FD:

$AB \rightarrow C$

Table 2

R2(A, D, G, H)

FD:

$AD \rightarrow GH$

Table 3

R3(B, D, E, F)

FD:

$BD \rightarrow EF$

Table 4

R4(A, I)

FD:

$A \rightarrow I$

Table 5

We need to retain the candidate key:

R5(A, B, D)

This relation contains the candidate key ABD.

Final Decomposition:

R1(A,B,C)

R2(A,D,G,H)

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R3(B,D,E,F)

R4(A,I)

R5(A, B, D)

Number of new tables = 5

If R(A, B, C, D, E, F, G, H, I, J)

then $H \rightarrow J$ must also be decomposed:

R6(H, J)

In that case:

Candidate Key

ABD

Highest Normal Form

1NF

Prime Attributes

A, B, D

Non-Prime Attributes

C, E, F, G, H, I, J

Number of tables after decomposition

6

Question 9: Consider relation:

R(A, B, C, D, E)

Functional dependencies:

- $A \rightarrow B$
- $B \rightarrow E$
- $C \rightarrow D$

We need to find the candidate key, prime/non-prime attributes, highest normal form, decomposition, and number of tables.

Answer: RHS attributes are:

- B
- E
- D

So the attributes not appearing on RHS are:

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A and C

Therefore, A and C must be included in the candidate key.

$(AC)^+$

Initially:

$\{A, C\}$

Using $A \rightarrow B$:

$\rightarrow B$

Using $B \rightarrow E$:

$\rightarrow E$

Using $C \rightarrow D$:

$\rightarrow D$

Therefore:

$(AC)^+ = \{A, B, C, D, E\}$

Since the closure contains all attributes:

Candidate Key = $\{A, C\}$

Step 2: Prime and Non-Prime Attributes

Prime Attributes: A, C

Non-Prime Attributes: B, D, E

Attribute	Type
A	Prime
B	Non-Prime
C	Prime
D	Non-Prime
E	Non-Prime

Step 3: Determine Highest Normal Form

Candidate key:

AC

Now check the functional dependencies.

$A \rightarrow B$

A is only a part of the candidate key AC.

B is a non-prime attribute.

Therefore, $A \rightarrow B$ is a partial dependency.

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$C \rightarrow D$

C is also only a part of candidate key AC.

D is a non-prime attribute.

Therefore, $C \rightarrow D$ is also a partial dependency.

Because partial dependencies exist, the relation violates 2NF.

Therefore:

Highest Normal Form = 1NF

Step 4: Decomposition

We decompose the relation to remove the partial and transitive dependencies.

Table 1

R1(A, B)

Functional dependency:

$A \rightarrow B$

Table 2

R2(B, E)

Functional dependency:

$B \rightarrow E$

This removes the transitive dependency:

$A \rightarrow B \rightarrow E$

Table 3

R3(C, D)

Functional dependency:

$C \rightarrow D$

Table 4

R4(A, C)

This contains the original candidate key AC.

Final Decomposition

The relation is decomposed into:

R1(A,B)

R2(B,E)

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R3(C,D)

R4(A, C)

Number of new tables = 4

LEARNING OUTCOMES:

1. Learned to identify functional dependencies and candidate keys by calculating attribute closures for different relational schemas.
2. Gained understanding of prime and non-prime attributes and their role in determining the normal form of a relation.
3. Learned to identify 1NF, 2NF, 3NF, and BCNF by analyzing partial and transitive dependencies in relational databases.
4. Developed practical knowledge of decomposition by dividing relations into smaller tables to eliminate redundancy and dependency-related anomalies.
5. Enhanced understanding of database normalization by applying functional dependencies, candidate keys, and decomposition techniques to design efficient and well-structured relational databases.